

Experiment No. 6

Inorganic Qualitative Analysis (Salt no. 2)

Aim: Analyse one basic (cation) and one acidic (anion) radical from given inorganic salt.

Apparatus: Test tubes, test tube stand, spatula, glass rod, watch glass, beaker, burner, etc.

A. Preliminary tests: (General scheme)

Test	Observation	Inference
Nature	crystalline	Cl^- , Br^- , I^- , NO_3^- , SO_4^{2-} may be present.
Color of salt	white	Pb^{2+} , Al^{3+} NH_4^+ may be present
Solubility	Soluble in water	crystalline salt present.

B. Dry tests for basic radicals (Cations)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of salt in a clean, dry test tube and heat it strongly. Observe the color of the residue and gas evolved.	white infusible residue,	Al^{3+} , Zn^{2+} , Ba^{2+} , Ca^{2+} NH_4^+ may be present.
2. Test for NH_4^+ Salt + NaOH, heat gently, hold moist turmeric paper near the mouth of test tube.	No smell of NH_3 gas	NH_4^+ present.
3. Charcoal cavity test: Take the salt and solid Na_2CO_3 in the ratio 1:2 on watch glass, add few drops of water and prepare a paste, place this in charcoal cavity and heat it with a blow pipe in a reducing flame.	—	—
4. Cobalt Nitrate Test: If white infusible residue is obtained in charcoal cavity test, add two drops of cobalt nitrate solution on the white residue and heat in the blue flame.	—	—

5. Flame test: Clean a platinum wire with conc. HCl. Moisten a small quantity of the substance with concentrated HCl on watch glass. Dip the cleaned platinum wire in it and heat it in oxidizing flame (non - luminous) and observe the color of the flame.		
6. Borax bead test: [For colored salt only] Make a small loop at the end of a clean platinum wire heat it strongly and dip it in borax powder. Heat it again till a glossy bead is obtained. Now touch the hot bead with a small crystal of the salt and heat it again in oxidizing (blue) flame. Observe the color of the bead.		

C. Dry tests for acidic radicals (Anions)

Test	Observation	Inference
i. Action of dil HCl: Salt + dil HCl.	No gas	CO_3^{2-} , SO_3^{2-} , NO_2^- absent
ii. Action of conc. H_2SO_4: Salt + conc. H_2SO_4 + heat	No gas	Cl^- , Br^- , NO_3^- , I^- CH_3COO^- absent.
iii) Action of MnO_2 and conc. H_2SO_4: Salt + a pinch of MnO_2 + conc. H_2SO_4 and heat	No gas	Cl^- , Br^- , I^- , NO_3^- absent.
iv) Cu-foil test: (If brown gas is obtained in the above test then only perform this test) Salt + Cu-foil + conc. H_2SO_4 and heat strongly	No brown gas	NO_3^- absent.

Note: (If the anion is not detected in above dry test, then salt may contain SO_4^{2-} or PO_4^{3-})

D. Preparation of solution (Original Solution): Dissolve about 0.5 gm of salt in 20 mL of water in a 50 mL beaker. This is called original solution (O.S.)

E. Wet test for basic radical (cations)

Detection of groups

Test	Observations	Inference
1. O.S. + dil. HCl	No ppt	q.r. I absent
2. O.S. + dil. HCl + H ₂ S gas / H ₂ S water solution	No ppt	q.r. II absent
3. O.S. + NH ₄ Cl + NH ₄ OH till alkaline	white ppt	q.r. III A Present.
4. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + H ₂ S gas / H ₂ S water solution	—	—
5. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + saturated solution of (NH ₄) ₂ CO ₃	—	—
6. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + Na ₂ HPO ₄	—	—
7. If all the above groups are absent	—	—

Analysis of Group: After detection of a particular group do analysis of that group.

Analysis of Group III A (Al³⁺ / Fe³⁺)

Test	Observations	Inference
O.S. + NH ₄ Cl + NH ₄ OH + ill alkaline	white gelatinous ppt	Al ³⁺ Present.
C.T. for Al³⁺ 1. O.S. + NaOH	white ppt Soluble in excess of NaOH & reobtained by adding NH ₄ Cl.	Al ³⁺ confirmed.
2. O.S. + Na ₂ HPO ₄	white ppt	Al ³⁺ confirmed.

F. Wet test for acidic radicals:

Detection of acidic radical from water solution:

1. Wet tests for CO_3^{2-} , SO_3^{2-} , Cl^- , Br^- , I^-

1. O.S. + AgNO_3		
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Distinction between halides (Cl^- , Br^- , I^-):

Note: (Perform this test only if ppt is obtained for AgNO_3 which is insoluble in dil. HNO_3)

Test	Observation	Inference
O.S. + Chloroform + Cl_2 water in excess (shake well)		

Note: (If one of the halide radical is present then further analysis is not necessary. Perform confirmatory test for that detected radical.)

2. Wet test for SO_4^{2-} and PO_4^{3-}

(This test is required to perform only, if ppt is NOT obtained in AgNO_3 test)

Test	Observation	Inference
O.S. + $\text{Ba}(\text{NO}_3)_2$	white ppt insoluble in dil. HNO_3	SO_4^{2-} present.

3. WET TEST FOR NO_2^-

Test	Observation	Inference
O.S. + dil. H_2SO_4 + FeSO_4 solution		

4. WET TEST FOR NO_3^-

Test	Observation	Inference
O.S. + 3-4 drops of conc. H_2SO_4 + 2-3 drops of diphenyl amine		

5. WET TEST FOR CH_3COO^-

Test	Observation	Inference
O.S. + FeCl_3 solution		

6. WET TEST FOR $\text{C}_2\text{O}_4^{2-}$

Test	Observation	Inference
O.S. + dil. Acetic acid + CaCl_2	—	—

C.T. for

Test	Observation	Inference
1. O.S. + Acetic acid + lead acetate solution	white ppt insoluble in dil. HNO_3 but soluble in hot ammonium acetate.	SO_4^{2-} confirmed.
2. O.S. + BaCl_2 solution	white ppt of BaSO_4	SO_4^{2-} confirmed.

Result:- Given inorganic salt no. 2 contains following

1. Basic radical Al^{3+}

2. Acidic radical SO_4^{2-}

3. Molecular formula of the salt $\text{Al}_2(\text{SO}_4)_3$

4. Hence the name of the salt Aluminium sulphate.

Remark and sign of teacher: